MATHEMATICS / INDEPENDENT STUDY

MM7033

Advanced Algebra

Progressive hints

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 2 · 25 October 2026

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

1 Unit 1 · Fields and splitting fields

MM7033-U1.01 · E · 15 min
Learning objective: U1.1 Minimal polynomials
Find the minimal polynomial and degree of $\sqrt{11}$ over $\mathbb{Q}$.
Hint 1 — a starting point. Use a monic quadratic.
Hint 2 — the key move. Eisenstein applies at eleven.

MM7033-U1.02 · E · 15 min
Learning objective: U1.1 Splitting fields
Find a splitting field of $x^2 - 3$ over $\mathbb{Q}$.
Hint 1 — a starting point. Both roots differ by a sign.
Hint 2 — the key move. Adjoining either root supplies the other.

MM7033-U1.03 · E · 15 min
Learning objective: U1.2 Finite-field subfields
List the subfield sizes in $\mathbb{F}_{3^4}$.
Hint 1 — a starting point. Use divisors of four.
Hint 2 — the key move. The degrees are one, two, and four.

MM7033-U1.04 · E · 15 min
Learning objective: U1.2 Separability
Show $x^5 - x$ over $\mathbb{F}_5$ has no repeated roots.
Hint 1 — a starting point. Compute the formal derivative in characteristic five.
Hint 2 — the key move. It is negative one.

MM7033-U1.05 · C · 30 min
Learning objective: U1.1 Splitting-field degree
Find the degree of the splitting field of $x^3 - 3$ over $\mathbb{Q}$.
Hint 1 — a starting point. Adjoin a real cube root and a primitive cube root of unity.
Hint 2 — the key move. The first degree is three, and the second is two because the first field is real.

MM7033-U1.06 · C · 30 min
Learning objective: U1.1 Embeddings
Describe all $\mathbb{Q}$-embeddings of $\mathbb{Q}\left(\sqrt{2}\right)$ into $\mathbb{C}$.
Hint 1 — a starting point. The image of the generator must satisfy its minimal polynomial.
Hint 2 — the key move. There are two choices of root.

MM7033-U1.07 · C · 30 min
Learning objective: U1.2 Frobenius order

Prove Frobenius on $\mathbb{F}_{p^n}$ has order exactly n.
Hint 1 — a starting point. Its nth power is the identity.
Hint 2 — the key move. A smaller positive power would make all elements roots of a polynomial of degree less than p^n.

MM7033-U1.08 · C · 30 min
Learning objective: U1.2 Derivatives

Prove a polynomial over a field of characteristic p has derivative zero exactly when all nonzero terms have exponents divisible by p.
Hint 1 — a starting point. Different monomials in the derivative cannot cancel each other's coefficients.
Hint 2 — the key move. The coefficient of x^{i-1} is ia_i.

MM7033-U1.09 · C · 30 min
Learning objective: U1.1 Changing the base

Compare the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ and over $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$.
Hint 1 — a starting point. The element already belongs to the larger base.
Hint 2 — the key move. Its degree over that base is one.

MM7033-U1.10 · A · 50 min
Learning objective: U1.1 Quartic splitting field

Prove the splitting field of $x^4 - 2$ over $\mathbb{Q}$ has degree eight.
Hint 1 — a starting point. Let α be the positive real fourth root of two.
Hint 2 — the key move. Its roots are $\pm\alpha, \pm i\alpha$.

MM7033-U1.11 · A · 50 min
Learning objective: U1.2 Subfield intersections

Inside $\mathbb{F}_{p^{12}}$, determine the intersection of its subfields of orders p^4 and p^6.
Hint 1 — a starting point. The intersection degree divides both four and six.
Hint 2 — the key move. Both subfields contain the unique field of order p^2.

MM7033-U1.12 · A · 50 min
Learning objective: U1.2 Inseparable irreducibles

Prove $x^p - t$ is irreducible and inseparable over $\mathbb{F}_{p(t)}$.
Hint 1 — a starting point. View it first over the UFD $\mathbb{F}_{p[t]}$.
Hint 2 — the key move. Use Eisenstein at the prime element t, then the derivative.

MM7033-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: Degrees inside a finite field

How many elements of $\mathbb{F}_{64}$ have minimal polynomial of degree six over $\mathbb{F}_2$? Justify that every smaller possible degree has been accounted for.

Hint 1 — a starting point. An element generates a subfield, and its degree divides six.

Hint 2 — the key move. The proper subfields have four and eight elements and intersect in the two-element field.

2 Unit 2 · Existence and transcendence

MM7033-U2.01 · E · 15 min
Learning objective: U2.1 Chains

Give a partially ordered set with two maximal elements and no greatest element.
Hint 1 — a starting point. Use incomparable elements.
Hint 2 — the key move. A two-element set with only equality as its order suffices.

MM7033-U2.02 · E · 15 min
Learning objective: U2.1 Proper ideals

Prove an ideal I of a unital ring is proper exactly when $1 \notin I$.
Hint 1 — a starting point. If one belongs, multiply it by every ring element.
Hint 2 — the key move. The converse follows because the whole ring contains one.

MM7033-U2.03 · E · 15 min
Learning objective: U2.2 Independence

Is $\{t, t^2\}$ algebraically independent over K in $K(t)$?
Hint 1 — a starting point. Look for a polynomial in two variables.
Hint 2 — the key move. Use $Y - X^2$.

MM7033-U2.04 · E · 15 min
Learning objective: U2.2 Transcendence bases

Give a transcendence basis of an algebraic extension L/K.
Hint 1 — a starting point. The extension already is algebraic over the base.
Hint 2 — the key move. The empty set is independent.

MM7033-U2.05 · C · 30 min
Learning objective: U2.1 Chain unions

Prove the union of a nonempty chain of proper ideals is proper and an ideal.
Hint 1 — a starting point. Place any two chosen elements in a common member using comparability.
Hint 2 — the key move. If one were in the union, it would be in one member.

MM7033-U2.06 · C · 30 min
Learning objective: U2.1 Maximal ideals over an ideal

Use maximal-ideal existence in a quotient to show every proper ideal I lies in a maximal ideal.
Hint 1 — a starting point. The quotient by a proper ideal is nonzero.
Hint 2 — the key move. Pull back a maximal ideal under the quotient map.

MM7033-U2.07 · C · 30 min
Learning objective: U2.2 Algebraic closure

Show an algebraically closed field has no proper algebraic field extension.
Hint 1 — a starting point. Use the minimal polynomial of any element in the extension.
Hint 2 — the key move. An irreducible polynomial over an algebraically closed field is linear.

MM7033-U2.08 · C · 30 min
Learning objective: U2.2 Transitivity

Prove algebraicity is transitive through a tower, allowing infinite extensions.
Hint 1 — a starting point. Only finitely many coefficients occur in an equation for one element.
Hint 2 — the key move. Put those coefficients into a finite intermediate extension.

MM7033-U2.09 · C · 30 min
Learning objective: U2.1 Extending independent sets

Use Zorn's lemma to extend a linearly independent subset of a vector space to a basis.
Hint 1 — a starting point. Order independent supersets of the given subset.
Hint 2 — the key move. A finite relation in a chain union lies in one chain member.

MM7033-U2.10 · A · 50 min
Learning objective: U2.2 A nontrivial transcendence basis

Show $\{t^2\}$ is a transcendence basis of $K(t)/K$.
Hint 1 — a starting point. A polynomial relation in t^2 would be a polynomial relation in t.
Hint 2 — the key move. The element t satisfies $X^2 - t^2$ over $K(t^2)$.

MM7033-U2.11 · A · 50 min
Learning objective: U2.1 Units detected by maximal ideals

Prove an element of a nonzero commutative unital ring is a unit exactly when it belongs to no maximal ideal.
Hint 1 — a starting point. A nonunit generates a proper ideal.
Hint 2 — the key move. Extend that ideal to a maximal ideal.

MM7033-U2.12 · A · 50 min
Learning objective: U2.2 Maximal independence

Prove every maximal algebraically independent subset B of L/K is a transcendence basis.
Hint 1 — a starting point. Assume an element is transcendental over $K(B)$.
Hint 2 — the key move. A relation after adjoining it would be a polynomial in that element with coefficients in $K(B)$.

MM7033-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Maximal ideals of eventually constant sequences

Let R be the ring of eventually constant real sequences, with coordinatewise operations. Classify its maximal ideals and show that one of them is not finitely generated.

Hint 1 — a starting point. Use coordinate evaluation and the eventual constant value as homomorphisms to $\mathbb{R}$.

Hint 2 — the key move. Let e_n be one at coordinate n and zero elsewhere; in a field an idempotent is zero or one.

3 Unit 3 · Modules and quotient constructions

MM7033-U3.01 · E · 15 min
Learning objective: U3.1 Annihilators

Compute the annihilator of $[4]$ in the $\mathbb{Z}$-module $\mathbb{Z}/12\mathbb{Z}$.
Hint 1 — a starting point. Solve $4r = 0$ modulo twelve.
Hint 2 — the key move. This holds exactly when three divides r.

MM7033-U3.02 · E · 15 min
Learning objective: U3.1 Module maps

Describe all $\mathbb{Z}$-module maps $\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$.
Hint 1 — a starting point. The image of one determines the map.
Hint 2 — the key move. Any of the five residue classes can be chosen.

MM7033-U3.03 · E · 15 min
Learning objective: U3.2 Submodules

Show an ideal of R is exactly a submodule of R as an R-module.
Hint 1 — a starting point. Compare the definitions.
Hint 2 — the key move. Both require an additive subgroup closed under multiplication by every scalar.

MM7033-U3.04 · E · 15 min
Learning objective: U3.2 Quotients

Describe the quotient $\frac{\mathbb{Z}^2}{N}$ for $N = \{(a, 0) : a \in \mathbb{Z}\}$.
Hint 1 — a starting point. Project to the second coordinate.
Hint 2 — the key move. The projection is onto with kernel N.

MM7033-U3.05 · C · 30 min
Learning objective: U3.1 Cyclic homomorphisms

Compute the number of $\mathbb{Z}$-module maps $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$.
Hint 1 — a starting point. The image of one must be annihilated by m.
Hint 2 — the key move. Count residues a with $n|ma$.

MM7033-U3.06 · C · 30 min
Learning objective: U3.1 Operator modules

Let T on K^2 satisfy $T(e_1) = 0, T(e_2) = e_1$. Show its $K[x]$-module is cyclic and identify it as a quotient.
Hint 1 — a starting point. The vector e_2 generates e_1 under x.
Hint 2 — the key move. The annihilator of e_2 is (x^2).

MM7033-U3.07 · C · 30 min
Learning objective: U3.2 Isomorphism theorem
In $\mathbb{Z}$, take submodules $A = 4\mathbb{Z}$ and $B = 6\mathbb{Z}$. Identify $(A + B)/B$ and $A/(A \cap B)$.
Hint 1 — a starting point. Compute gcd and lcm ideals.
Hint 2 — the key move. The sum is $2\mathbb{Z}$ and intersection is $12\mathbb{Z}$.

MM7033-U3.08 · C · 30 min
Learning objective: U3.2 Simple modules
Classify the simple $\mathbb{Z}$-modules up to isomorphism.
Hint 1 — a starting point. Use maximal ideals of $\mathbb{Z}$.
Hint 2 — the key move. They are generated by primes.

MM7033-U3.09 · C · 30 min
Learning objective: U3.1 Torsion
For a domain R, prove the torsion elements of an R-module form a submodule.
Hint 1 — a starting point. Choose a nonzero annihilator for each of two elements.
Hint 2 — the key move. The product of the annihilators is still nonzero.

MM7033-U3.10 · A · 50 min
Learning objective: U3.1 Maps from a quotient
Prove naturally that $\mathrm{Hom}_{R(R/I,M)}$ is isomorphic to $\{m \in M : Im = 0\}$ as an R-module.
Hint 1 — a starting point. Evaluate a map at $[1]$.
Hint 2 — the key move. Reconstruct by $[r] \mapsto rm$.

MM7033-U3.11 · A · 50 min
Learning objective: U3.2 Splitting a torsion module
Prove $\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ as modules, and identify the images of a generator.
Hint 1 — a starting point. Use simultaneous reduction.
Hint 2 — the key move. Coprimality gives surjectivity by Bézout.

MM7033-U3.12 · A · 50 min
Learning objective: U3.1 Submodule invariance
For a vector space with operator T, prove its $K[x]$-submodules are exactly its T-invariant vector subspaces.
Hint 1 — a starting point. Constants in $K[x]$ supply scalar multiplication.
Hint 2 — the key move. Closure under T gives closure under every polynomial in T.

MM7033-U3.13 · A · 60 min · Optional challenge
Learning objective: U3 Transfer: A single nilpotent chain
On K^3 let $T(e_1) = 0$, $T(e_2) = e_1$, and $T(e_3) = e_2$. Determine all T-invariant vector subspaces, with proof that your list is exhaustive.
Hint 1 — a starting point. Choose a vector in an invariant subspace with its highest nonzero coordinate as far to the right as possible.
Hint 2 — the key move. Apply powers of T, then subtract the lower coordinate terms.

4 Unit 4 · Free modules and exact sequences

MM7033-U4.01 · E · 15 min
Learning objective: U4.1 Presentations
Give a one-generator, one-relation presentation of $\mathbb{Z}/9\mathbb{Z}$.
Hint 1 — a starting point. Use multiplication by nine.
Hint 2 — the key move. Its image is exactly the kernel of reduction.

MM7033-U4.02 · E · 15 min
Learning objective: U4.2 Exactness
Check exactness of $0 \to K \xrightarrow{i} K^2 \xrightarrow{p} K \to 0$, where $i(a) = (a, 0)$ and $p(a, b) = b$.
Hint 1 — a starting point. Compute kernel and image.
Hint 2 — the key move. Both the kernel of p and image of i are the first coordinate axis.

MM7033-U4.03 · E · 15 min
Learning objective: U4.1 Finite support
Give an element of $\prod_{n\geq 0} \mathbb{Z}$ not lying in $\bigoplus_{n\geq 0} \mathbb{Z}$.
Hint 1 — a starting point. Use infinitely many nonzero coordinates.
Hint 2 — the key move. The all-ones sequence works.

MM7033-U4.04 · E · 15 min
Learning objective: U4.1 Free maps
How is a linear map $R^3 \to M$ determined?
Hint 1 — a starting point. Choose the images of the three standard basis elements.
Hint 2 — the key move. Extend by scalar combinations.

MM7033-U4.05 · C · 30 min
Learning objective: U4.2 Nonsplitting
Prove $0 \to \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$ does not split.
Hint 1 — a starting point. A section's image of $[1]$ must be annihilated by three.
Hint 2 — the key move. No nonzero integer is annihilated by three.

MM7033-U4.06 · C · 30 min
Learning objective: U4.2 Free quotients
Show a short exact sequence with quotient R^n splits.
Hint 1 — a starting point. Lift each standard basis vector.
Hint 2 — the key move. Extend the chosen lifts linearly.

MM7033-U4.07 · C · 30 min
Learning objective: U4.1 Presentation computation
Identify the $\mathbb{Z}$-module generated by u, v with relations $u - 2v = 0$ and $6v = 0$.
Hint 1 — a starting point. Eliminate u using the first relation.
Hint 2 — the key move. The remaining generator has order six.

MM7033-U4.08 · C · 30 min
Learning objective: U4.2 Direct summands
Prove a direct summand of a projective module is projective.
Hint 1 — a starting point. Extend a map from the summand using the projection.
Hint 2 — the key move. Lift through the larger module and restrict back.

MM7033-U4.09 · C · 30 min
Learning objective: U4.1 Finite rank
Prove R^2 cannot be isomorphic to R^3 for a nonzero commutative unital ring.
Hint 1 — a starting point. Reduce modulo a maximal ideal.
Hint 2 — the key move. An isomorphism induces an isomorphism of residue-field vector spaces.

MM7033-U4.10 · A · 50 min
Learning objective: U4.2 Projective cyclic example
Show $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$-module.
Hint 1 — a starting point. Apply its lifting property to reduction $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$.
Hint 2 — the key move. Lifting the identity would provide a section.

MM7033-U4.11 · A · 50 min
Learning objective: U4.2 A projective nonfree module
Let $R = K \times K$ and $e = (1, 0)$. Prove Re is projective but is not free.
Hint 1 — a starting point. Use $R = Re \oplus R(1 - e)$.
Hint 2 — the key move. Every element of Re is annihilated by $(0, 1)$.

MM7033-U4.12 · A · 50 min
Learning objective: U4.2 Splitting from a retraction
Given a short exact sequence and a retraction $r : B \to A$ of its injection i, construct the isomorphism $B \cong A \oplus C$.
Hint 1 — a starting point. Subtract $i(r(b))$ to isolate the complementary part.
Hint 2 — the key move. The restriction of the quotient map to $\ker r$ is an isomorphism.

MM7033-U4.13 · A · 60 min · Optional challenge

Learning objective: U4 Transfer: Lifting maps from a residue ideal

Let $R = \mathbb{Z}/6\mathbb{Z}$ and $M = 2R$. Given any surjective R-linear map $q : N \to P$ and any R-linear map $f : M \to P$, construct an R-linear $h : M \to N$ with $qh = f$. Is M free over R? Justify your answer.

Hint 1 — a starting point. Find an element of M that acts as the identity on every element of M.

Hint 2 — the key move. The residue e of four satisfies e squared equals e and M equals Re. Lift f(e), then multiply the lift by e.

5 Unit 5 · Tensor products

MM7033-U5.01 · E · 15 min
Learning objective: U5.1 Tensor relations
Simplify $(2m) \otimes n + m \otimes (3n)$ over $\mathbb{Z}$.
Hint 1 — a starting point. Move scalars to the tensor.
Hint 2 — the key move. Both terms are multiples of $m \otimes n$.

MM7033-U5.02 · E · 15 min
Learning objective: U5.2 Cyclic tensors
Compute $\mathbb{Z}/8\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}$.
Hint 1 — a starting point. Use the gcd formula.
Hint 2 — the key move. The gcd is four.

MM7033-U5.03 · E · 15 min
Learning objective: U5.1 Unit tensor
Give inverse maps proving $R \otimes_R M \cong M$.
Hint 1 — a starting point. Multiply the scalar into the module element.
Hint 2 — the key move. Use $m \mapsto 1 \otimes m$ for the inverse.

MM7033-U5.04 · E · 15 min
Learning objective: U5.2 Tensor annihilation
Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/7\mathbb{Z}$.
Hint 1 — a starting point. Use the quotient-ring tensor identity.
Hint 2 — the key move. Multiplication by seven is onto on $\mathbb{Q}$.

MM7033-U5.05 · C · 30 min
Learning objective: U5.1 Pure tensors
Show $e_1 \otimes e_1 + e_2 \otimes e_2$ in $K^2 \otimes K^2$ is not a pure tensor.
Hint 1 — a starting point. Compare its four basis coefficients with those of $u \otimes v$.
Hint 2 — the key move. A pure tensor gives a matrix of rank at most one.

MM7033-U5.06 · C · 30 min
Learning objective: U5.1 Tensor basis
Prove $R^m \otimes_R R^n$ is free of rank mn.
Hint 1 — a starting point. Expand tensors in the standard bases.
Hint 2 — the key move. Coordinate-product bilinear maps detect each coefficient.

MM7033-U5.07 · C · 30 min
Learning objective: U5.2 Quotient tensors
Compute $R/I \otimes_R R/J$.
Hint 1 — a starting point. Apply the quotient formula to $M = R/J$.
Hint 2 — the key move. The resulting quotient kills both ideals.

MM7033-U5.08 · C · 30 min
Learning objective: U5.2 Failure of injectivity
Tensor $0 \to \mathbb{Z} \xrightarrow{\times 4} \mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to 0$ with $\mathbb{Z}/2\mathbb{Z}$. Describe every resulting map.
Hint 1 — a starting point. Identify each nonzero tensor using the gcd formula.
Hint 2 — the key move. The first induced map is multiplication by four modulo two.

MM7033-U5.09 · C · 30 min
Learning objective: U5.1 Distributivity
Prove $(M \oplus N) \otimes P \cong (M \otimes P) \oplus (N \otimes P)$.
Hint 1 — a starting point. Send a pair tensor to its two coordinate tensors.
Hint 2 — the key move. Reverse the map using the two inclusions.

MM7033-U5.10 · A · 50 min
Learning objective: U5.2 Right exactness
For a submodule $K \leq M$, identify the kernel of $M \otimes N \to (M/K) \otimes N$.
Hint 1 — a starting point. The quotient imposes relations $k \otimes n = 0$.
Hint 2 — the key move. Do not assume the map $K \otimes N \to M \otimes N$ is injective.

MM7033-U5.11 · A · 50 min
Learning objective: U5.1 Tensor–Hom correspondence
Prove $\mathrm{Hom}_{R(M \otimes N, P)} \cong \mathrm{Hom}_{R\left(M, \mathrm{Hom}_{R(N,P)}\right)}$ naturally.
Hint 1 — a starting point. Curry a bilinear map.
Hint 2 — the key move. A map $m \mapsto (n \mapsto b(m,n))$ is linear exactly when b is bilinear.

MM7033-U5.12 · A · 50 min
Learning objective: U5.2 A torsion tensor vanishes
Prove $(\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}(\mathbb{Q}/\mathbb{Z})} = 0$.
Hint 1 — a starting point. Every element in the first factor is killed by some positive integer.
Hint 2 — the key move. The second factor is divisible by that same integer.

MM7033-U5.13 · A · 60 min · Optional challenge

Learning objective: U5 Transfer: Relations after changing coefficients

Let M be generated over $\mathbb{Z}$ by u,v with relations $2u + 3v = 0$ and $4u + 9v = 0$. Compute $M \otimes_{\mathbb{Z}} \mathbb{Q}$, $M \otimes_{\mathbb{Z}} \mathbb{F}_2$, and $M \otimes_{\mathbb{Z}} \mathbb{F}_3$. Decide whether M is zero.

Hint 1 — a starting point. Tensor the presentation and compute each resulting cokernel.

Hint 2 — the key move. The two relation vectors are the columns $(2, 3)$ and $(4, 9)$.

6 Unit 6 · Localization

MM7033-U6.01 · E · 15 min
Learning objective: U6.1 Localized integers
Describe $\mathbb{Z}\left[\frac{1}{3}\right]$ as a subring of $\mathbb{Q}$.
Hint 1 — a starting point. The allowed denominators are powers of three.
Hint 2 — the key move. Fractions may simplify but need not have a unique presentation.

MM7033-U6.02 · E · 15 min
Learning objective: U6.1 Units at a prime
Which of $2, 5, 10$ are units in $\mathbb{Z}_{(5)}$?
Hint 1 — a starting point. Denominators not divisible by five are inverted.
Hint 2 — the key move. An integer is a unit exactly when five does not divide it.

MM7033-U6.03 · E · 15 min
Learning objective: U6.1 Vanishing
In $\mathbb{Z}/10\mathbb{Z}$, localize at powers of two. Show the class of five becomes zero.
Hint 1 — a starting point. Use the localization vanishing criterion.
Hint 2 — the key move. Two times five is zero in the original ring.

MM7033-U6.04 · E · 15 min
Learning objective: U6.2 Localizing a cyclic module
Compute $\mathbb{Z}\left[\frac{1}{2}\right] \otimes_{\mathbb{Z}} \mathbb{Z}/8\mathbb{Z}$.
Hint 1 — a starting point. Use quotient by eight times the localized ring.
Hint 2 — the key move. Eight is now a unit.

MM7033-U6.05 · C · 30 min
Learning objective: U6.1 Injectivity criterion
Prove the map $R \to S^{-1}R$ is injective exactly when no element of S annihilates a nonzero element of R.
Hint 1 — a starting point. Compute its kernel by the vanishing criterion.
Hint 2 — the key move. An element maps to zero iff some denominator-set element kills it.

MM7033-U6.06 · C · 30 min
Learning objective: U6.1 Identifying a localization
Prove localizing $\mathbb{Z}/6\mathbb{Z}$ at powers of two gives $\mathbb{F}_3$.
Hint 1 — a starting point. Reduction modulo three inverts two.
Hint 2 — the key move. The kernel that disappears is generated by three.

MM7033-U6.07 · C · 30 min
Learning objective: U6.2 A common annihilator
If M is finitely generated and $S^{-1}M = 0$, prove some $s \in S$ kills all of M.
Hint 1 — a starting point. Choose one annihilator for each generator.
Hint 2 — the key move. Multiply the finitely many annihilators.

MM7033-U6.08 · C · 30 min
Learning objective: U6.2 Quotients and localization
Show $S^{-1}(M/N) \cong S^{-1}M/\operatorname{im}(S^{-1}N)$.
Hint 1 — a starting point. Localize the quotient exact sequence.
Hint 2 — the key move. Exactness identifies the cokernel.

MM7033-U6.09 · C · 30 min
Learning objective: U6.1 Local rings
Show the nonunits in $R_{\mathfrak{p}}$ form its unique maximal ideal.
Hint 1 — a starting point. Fractions with numerator outside the prime are invertible.
Hint 2 — the key move. Fractions with numerator inside it form a proper ideal.

MM7033-U6.10 · A · 50 min
Learning objective: U6.2 Detecting zero locally
Prove an R-module M is zero if $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$. No finite-generation hypothesis is allowed.
Hint 1 — a starting point. If $m \neq 0$, its annihilator is a proper ideal.
Hint 2 — the key move. Choose a maximal ideal containing that annihilator.

MM7033-U6.11 · A · 50 min
Learning objective: U6.1 Localization in stages
Show inverting f and then g in a commutative ring is equivalent to inverting fg.
Hint 1 — a starting point. Compare the maps each construction permits into a target.
Hint 2 — the key move. If fg is a unit, both f and g are units.

MM7033-U6.12 · A · 50 min
Learning objective: U6.2 Flatness of localization (proof reconstruction)
Prove $S^{-1}R$ is flat by explicitly checking that an injection remains injective after localization and identifying localization with tensoring.
Hint 1 — a starting point. Use a vanishing fraction in the image.
Hint 2 — the key move. An element of S killing the image must kill the original element by injectivity.

MM7033-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: Two localized views of a crossing

Let $A = K[x,y]/(xy)$. Consider the map $A \to K[x,x^{-1}] \times K[y,y^{-1}]$ obtained by setting y=0 in the first coordinate and x=0 in the second. Is it injective or surjective? Describe its image exactly.

Hint 1 — a starting point. Reduce a polynomial modulo xy before evaluating either coordinate.

Hint 2 — the key move. A reduced expression has one constant term, positive powers of x, and positive powers of y.

7 Unit 7 · Multilinear algebra

MM7033-U7.01 · E · 15 min
Learning objective: U7.1 Exterior arithmetic
Expand $(e_1 + 2e_2) \wedge (3e_1 + e_2)$ in $\wedge^2 R^2$.
Hint 1 — a starting point. Expand bilinearly and eliminate repeated factors.
Hint 2 — the key move. Use $e_2 \wedge e_1 = -e_1 \wedge e_2$.

MM7033-U7.02 · E · 15 min
Learning objective: U7.1 Exterior rank
Give a basis and rank of $\wedge^2 R^4$.
Hint 1 — a starting point. Choose increasing pairs of basis indices.
Hint 2 — the key move. There are six pairs.

MM7033-U7.03 · E · 15 min
Learning objective: U7.2 Trace and determinant
Find trace and determinant of $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ over $\mathbb{Z}$.
Hint 1 — a starting point. Use diagonal sum and the two-by-two determinant.
Hint 2 — the key move. The off-diagonal term contributes no determinant product here.

MM7033-U7.04 · E · 15 min
Learning objective: U7.1 Repeated inputs
Why is $v \wedge v = 0$ even in characteristic two?
Hint 1 — a starting point. Use the definition of the quotient.
Hint 2 — the key move. It is an imposed alternating relation.

MM7033-U7.05 · C · 30 min
Learning objective: U7.2 Cyclic trace
Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for square matrices over a commutative ring.
Hint 1 — a starting point. Expand diagonal entries as sums.
Hint 2 — the key move. Interchange the two indices and commute scalar factors.

MM7033-U7.06 · C · 30 min
Learning objective: U7.2 Similarity invariance
Prove similar matrices have equal determinant and trace.
Hint 1 — a starting point. Write the similar matrix as $P^{-1}AP$.
Hint 2 — the key move. Use determinant multiplicativity and cyclicity of trace.

MM7033-U7.07 · C · 30 min
Learning objective: U7.1 Functorial exterior powers
Show $\wedge^{n(TS)} = (\wedge^n T)(\wedge^n S)$.
Hint 1 — a starting point. Evaluate on a generating pure wedge.
Hint 2 — the key move. Both sides apply TS to each input.

MM7033-U7.08 · C · 30 min
Learning objective: U7.2 Rank-one endomorphisms
Let $T : M \to M$ have form $T(m) = \varphi(m)v$ on a finite free module. Prove $\operatorname{tr}(T) = \varphi(v)$.
Hint 1 — a starting point. Use a basis and its dual.
Hint 2 — the key move. Expand v in that basis.

MM7033-U7.09 · C · 30 min
Learning objective: U7.1 Vanishing top powers
Prove $\wedge^n R^r = 0$ for $n > r$.
Hint 1 — a starting point. Expand in the standard basis.
Hint 2 — the key move. Every n-tuple of basis indices repeats an index.

MM7033-U7.10 · A · 50 min
Learning objective: U7.2 Dual bases
For a finite projective module, prove the trace formula is independent of the chosen dual basis.
Hint 1 — a starting point. Expand one basis family using the other.
Hint 2 — the key move. Use the reconstruction identity to collapse the double sum.

MM7033-U7.11 · A · 50 min
Learning objective: U7.1 Exterior direct-sum decomposition
For finite free modules M, N, prove $\wedge^2 (M \oplus N) \cong \wedge^2 M \oplus (M \otimes N) \oplus \wedge^2 N$.
Hint 1 — a starting point. Use bases grouped by the two summands.
Hint 2 — the key move. Sorted pairs are of three types: both in M, one in each, or both in N.

MM7033-U7.12 · A · 50 min
Learning objective: U7.2 Finite-order operators
Let $T^5 = 1$ on a finite-dimensional vector space. Show $\det(T)^5 = 1$, and give a reason this does not force T to be the identity.
Hint 1 — a starting point. Apply determinant to the power equation.
Hint 2 — the key move. A nontrivial fifth-root scalar operator is a counterexample over $\mathbb{C}$.

MM7033-U7.13 · A · 60 min · Optional challenge
Learning objective: U7 Transfer: The determinant of a projection
Let $P^2 = P$ on a finite-dimensional K-vector space and let r be its rank. Prove $\det(I + tP) = (1 + t)^r$ in $K[t]$. Does $\mathrm{tr}(P) = 0$ force $P = 0$?
Hint 1 — a starting point. Separate a vector into Pv and v-Pv.
Hint 2 — the key move. Use bases of the image and kernel; examine what happens to the integer r in positive characteristic.

8 Unit 8 · Noetherian methods and applications

MM7033-U8.01 · E · 15 min
Learning objective: U8.1 Noetherian examples
Explain why every finite-dimensional vector space is a Noetherian module over its field.
Hint 1 — a starting point. An increasing chain cannot increase dimension indefinitely.
Hint 2 — the key move. Equal-dimensional nested subspaces are equal.

MM7033-U8.02 · E · 15 min
Learning objective: U8.1 Hilbert's theorem
Why is $K[x, y, z]$ Noetherian for a field K?
Hint 1 — a starting point. A field has only two ideals.
Hint 2 — the key move. Apply Hilbert's theorem three times.

MM7033-U8.03 · E · 15 min
Learning objective: U8.2 Coordinate rings
Show $K[x, y]/(xy)$ is not a domain.
Hint 1 — a starting point. Use the two coordinate classes.
Hint 2 — the key move. Neither class lies in the ideal, but their product does.

MM7033-U8.04 · E · 15 min
Learning objective: U8.2 Specialization
Compute $R^3 \otimes_R R/M$ for a maximal ideal M.
Hint 1 — a starting point. Tensor commutes with finite direct sums.
Hint 2 — the key move. Each copy of R becomes R/M.

MM7033-U8.05 · C · 30 min
Learning objective: U8.1 Quotients
Prove a quotient of a Noetherian module is Noetherian.
Hint 1 — a starting point. Lift an increasing chain through the quotient map.
Hint 2 — the key move. The preimages form an increasing chain of submodules.

MM7033-U8.06 · C · 30 min
Learning objective: U8.1 Finitely generated submodules
Prove the ascending chain condition implies every submodule is finitely generated.
Hint 1 — a starting point. Assume a counterexample and choose generators one at a time.
Hint 2 — the key move. At each finite stage choose an element outside the current span.

MM7033-U8.07 · C · 30 min
Learning objective: U8.1 Infinite polynomial ring
Show $K[x_1, x_2, ...]$ is not Noetherian.
Hint 1 — a starting point. Use ideals generated by initial segments of variables.
Hint 2 — the key move. The next variable does not belong to the preceding ideal.

MM7033-U8.08 · C · 30 min
Learning objective: U8.2 Local coordinate calculation
Compute the localization at x of $A = K[x, y]/(xy)$.
Hint 1 — a starting point. Inverting x forces y to vanish.
Hint 2 — the key move. The remaining coordinate is an invertible x.

MM7033-U8.09 · C · 30 min
Learning objective: U8.2 Finite presentation
Prove every finitely generated module over a Noetherian ring is finitely presented.
Hint 1 — a starting point. Choose a finite free module surjecting onto it.
Hint 2 — the key move. Its kernel is finitely generated.

MM7033-U8.10 · A · 50 min
Learning objective: U8.1 Localization preserves Noetherianity
Prove a localization of a Noetherian ring is Noetherian.
Hint 1 — a starting point. Contract an ideal of the localization back to the original ring.
Hint 2 — the key move. Every fraction in the ideal has its numerator in the contracted ideal after multiplying by a unit.

MM7033-U8.11 · A · 50 min
Learning objective: U8.2 Eigenspace decomposition
Suppose $T^2 = 1$ on a vector space over a field of characteristic not two. Construct projections onto the two eigenspaces and prove a direct-sum decomposition.
Hint 1 — a starting point. Use polynomials in T.
Hint 2 — the key move. Take $\frac{1+T}{2}$ and $\frac{1-T}{2}$.

MM7033-U8.12 · A · 50 min
Learning objective: U8.1 Proof reconstruction: Hilbert reduction step
In the proof of Hilbert's theorem, explain in detail why finitely many polynomials with leading coefficients generating the leading-coefficient ideal suffice together with finitely many bounded-degree polynomials.
Hint 1 — a starting point. Subtract shifted multiples to cancel the leading term.
Hint 2 — the key move. The bounded-degree intersection is a submodule of a finite free module.

MM7033-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: A subring with too many generators

Inside $K[x, y]$, let $B = K + xK[x, y]$. Prove B is a subring and is not Noetherian, although $K[x, y]$ is Noetherian.

Hint 1 — a starting point. Inspect ideals of B generated by x, xy, and further powers of y attached to x.

Hint 2 — the key move. Multiplying an element xy^j by an element of B changes its coefficient of x only by a scalar multiple of y^j.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7033-M1.01 · E · 15 min
Learning objective: U1 Finite fields
List the subfield sizes of $\mathbb{F}_{2^8}$ and justify that the list is complete.
Hint 1 — a starting point. Use divisors of eight.
Hint 2 — the key move. The allowed degrees are one, two, four, and eight.

MM7033-M1.02 · E · 15 min
Learning objective: U5 Tensor computation
Compute $\mathbb{Z}/15\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/21\mathbb{Z}$.
Hint 1 — a starting point. Use the sum of the two annihilator ideals.
Hint 2 — the key move. Their gcd is three.

MM7033-M1.03 · C · 30 min
Learning objective: U3–U4 Presentations
Identify the module generated by a, b with relations $a = 3b$ and $10b = 0$, and decide whether it is projective over $\mathbb{Z}$.
Hint 1 — a starting point. Eliminate a.
Hint 2 — the key move. A section of $\mathbb{Z} \to \mathbb{Z}/10\mathbb{Z}$ cannot exist.

MM7033-M1.04 · C · 30 min
Learning objective: U6 Localization
Compute $\mathbb{Z}\left[\frac{1}{3}\right] \otimes_{\mathbb{Z}} \mathbb{Z}/12\mathbb{Z}$.
Hint 1 — a starting point. Quotient the localized ring by twelve.
Hint 2 — the key move. Three is a unit, so the ideal generated by twelve equals that generated by four.

MM7033-M1.05 · A · 40 min
Learning objective: U2–U3 Simple quotients
Prove every nonzero finitely generated module over a commutative unital ring has a maximal proper submodule.
Hint 1 — a starting point. Apply Zorn to proper submodules.
Hint 2 — the key move. Finite generation prevents a chain union from becoming the whole module without one member doing so.

MM7033-M1.06 · A · 40 min
Learning objective: U7–U8 Operators
Let $T^3 = 1$ over $\mathbb{C}$. Prove the space is a direct sum of the three eigenspaces and describe projections using interpolation.
Hint 1 — a starting point. The roots of $t^3 - 1$ are distinct.
Hint 2 — the key move. Use polynomials equal to one at one root and zero at the others.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7033-M2.01 · E · 15 min
Learning objective: U3 Module maps
Compute $\mathrm{Hom}_{\mathbb{Z}(\mathbb{Z}/6\mathbb{Z},\mathbb{Z}/9\mathbb{Z})}$ as an abelian group.
Hint 1 — a starting point. A map is determined by the image of one.
Hint 2 — the key move. That image must be killed by six.

MM7033-M2.02 · E · 15 min
Learning objective: U7 Exterior powers
Compute the rank of $\wedge^3 R^5$ and explain why $\wedge^6 R^5 = 0$.
Hint 1 — a starting point. Count increasing index sets.
Hint 2 — the key move. Repeated basis indices force a wedge to vanish.

MM7033-M2.03 · C · 30 min
Learning objective: U1 Splitting fields
Find the degree of the splitting field of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.
Hint 1 — a starting point. Adjoin both square roots.
Hint 2 — the key move. Show the second is not in the first quadratic field.

MM7033-M2.04 · C · 30 min
Learning objective: U8 Noetherianity
Prove $K[x,y]/(x^2,xy)$ is Noetherian. Must it be a domain?
Hint 1 — a starting point. Use Hilbert and quotient preservation.
Hint 2 — the key move. The nonzero class of x has square zero.

MM7033-M2.05 · A · 40 min
Learning objective: U5–U6 Local detection of a map
Prove a module map is an isomorphism if its localization at every maximal ideal is an isomorphism.
Hint 1 — a starting point. Use its kernel and cokernel.
Hint 2 — the key move. Localization is exact, and zero modules can be detected at maximal ideals.

MM7033-M2.06 · A · 40 min
Learning objective: U4–U7 Projective trace
Let M be finite projective and $T(m)=\varphi(m)v$. Prove $\mathrm{tr}(T)=\varphi(v)$ using a dual basis.
Hint 1 — a starting point. Insert the rank-one formula into the trace definition.
Hint 2 — the key move. Apply φ to the dual-basis reconstruction of v.